

HIE Knowledge Retrieval Engine: Scalable Semantic ETL & RAG Platform

B.Tech CSE (AI) Final Year I

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An End-to-End Semantic ETL and RAG System for Electronic Health Records





The Problem: Fragmented Healthcare Data

Healthcare providers face significant challenges due to fragmented Electronic Health Record (EHR) data, leading to inefficiencies and potential risks.

Fragmented Clinical Data

Data scattered across various systems (Athena, Elation, etc.) creates a disjointed patient view, hindering comprehensive care.

Time-Consuming Chart Reviews

On average, providers spend **20 minutes** per patient manually reviewing charts, diverting precious time from direct patient interaction.

Provider Burnout

The administrative burden of managing disparate data sources contributes significantly to provider fatigue and burnout.

The Solution: Automated Semantic ETL + Local RAG

Elation Infiniwell Docsnap provides a robust, automated solution to integrate and interpret fragmented EHR data, transforming it into actionable insights.

Automated High-Throughput Semantic ETL

Our **Semantic ETL pipeline** ingests vast amounts of clinical data, normalizes it, and extracts meaningful information, setting the stage for intelligent retrieval.

- Efficient data ingestion from diverse EHRs
- Normalization via industry standards (LOINC/SNOMED)
- Structured data output for advanced analytics

Local RAG (Retrieval-Augmented Generation)

The **Local RAG pipeline** leverages advanced AI to provide relevant clinical intelligence rapidly, directly at the point of care.

- Context-aware data retrieval
- Privacy-first local inference
- Generates concise, summarized insights



Unlocking Value: Efficiency & Insight

The Elation Infiniwell Docsnap delivers tangible benefits, dramatically improving operational efficiency and clinical decision-making.

80%

Reduction

In manual data handling and chart review time, freeing up clinicians for patient care.

< 5

Seconds

For insight retrieval, providing critical information almost instantly to providers.

100%

Data Accuracy

Through rigorous semantic normalization and validation protocols.



Architectural Deep-Dive: System Overview

The HIE System is built on a modular, scalable architecture designed for high-throughput and reliability. Figure 1 illustrates the end-to-end pipeline.

Figure 1: High-Level Docsnap Pipeline Overview

This diagram showcases the ingestion points, processing engine, formatting layer, and various export mechanisms, highlighting the modularity of the pipeline.





Data Ingestion Layer: Connecting Diverse Sources

Our system supports a variety of ingestion methods to ensure compatibility with different EHR systems and data formats, providing a comprehensive data intake strategy.



API Polling (Elation/Athena)

Direct, real-time data retrieval from major EHR vendors like Elation and Athena via secure API connections.



SFTP Connectors

Secure File Transfer Protocol (SFTP) for batch processing and integration with legacy systems.



SQL Connectors

Direct database connections to extract structured data from various SQL-based repositories.



Manual Uploads

Support for manual document uploads (e.g., CCDA, HL7 files) for ad-hoc or smaller datasets.

Orchestration Engine:

hasa_alerts.py

The `hasa_alerts.py` script serves as the core orchestration engine, managing data processing workflows, concurrency, and task scheduling.

Threading Model

Utilizes a **multi-threaded approach** to parallelize data ingestion and processing, optimizing resource utilization and throughput.

- Manages concurrent data streams
- Distributes workload efficiently
- Ensures high performance under load

File-Locking Mechanism

A critical **file-locking mechanism** prevents race conditions and data corruption during concurrent file operations, maintaining data integrity.

- Ensures atomic file access
- Prevents simultaneous writes
- Guarantees data consistency





Semantic Transformation: From Raw Data to Knowledge

The transformation layer is crucial for converting disparate clinical data formats into a standardized, semantically rich representation, making it machine-readable and interoperable.

1

CCDA/HL7 Parsing

Ingestion of complex **CCDA** and **HL7 documents**, the backbone of health information exchange.

2

JSON Conversion

Conversion of parsed data into a standardized, lightweight **JSON format** for easier processing and storage.

3

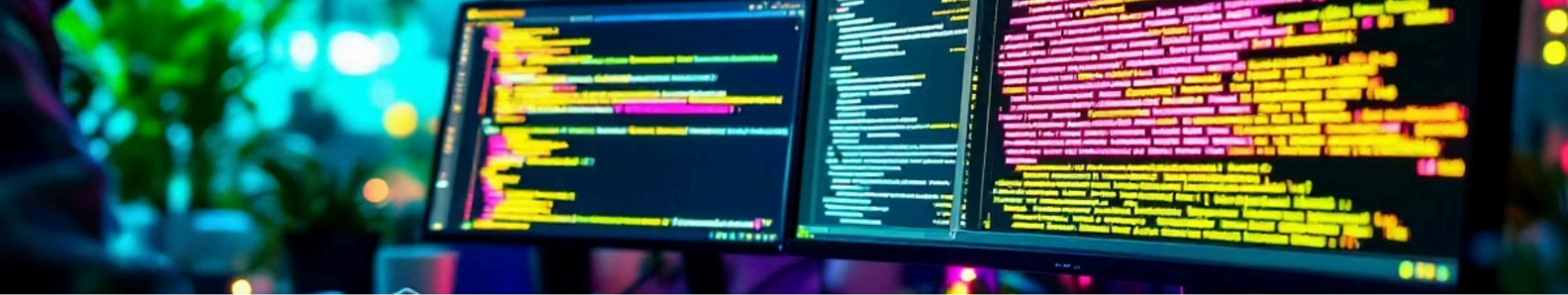
LOINC/SNOMED Mapping

Mapping of clinical concepts to industry-standard terminologies like **LOINC** (laboratory tests) semantic interoperability.

Multi-Channel Transport: `hasa_transports.py`

The `hasa_transports.py` module facilitates flexible and secure data export to various destinations, ensuring broad compatibility and accessibility.

SFTP	Secure File Transfer Protocol for pushing processed data.	Batch data delivery to partners.
S3	Amazon S3 integration for cloud storage and access.	Scalable cloud archiving and analytics.
Mirth	Integration with Mirth Connect for advanced HIE.	Real-time data routing in healthcare networks.
SMTP	Simple Mail Transfer Protocol for email notifications.	Alerts and notifications to stakeholders.



Configuration as Code: JSON Driven Workflows

Our "Configuration as Code" approach uses JSON files to define and manage critical operational parameters, ensuring flexibility, version control, and ease of deployment.

config/customers.json

Stores partner credentials and specific configurations for each customer integration, ensuring secure and tailored data exchange.

config/fieldset.json

Defines data extraction schemas, specifying which fields to extract and how they should be transformed from source documents.

config/csv_mapping.json

Manages header normalization for various CSV inputs, standardizing incoming data for consistent processing.

config/hasa_alerts_config.json

Contains orchestration logic, defining workflows, alert triggers, and processing rules for the engine.